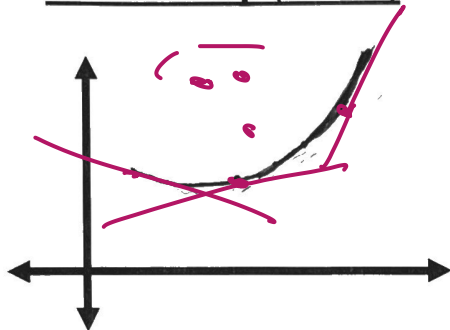


The Second Derivative Test for Concavity

Concavity

• Concave Up (C. U.)



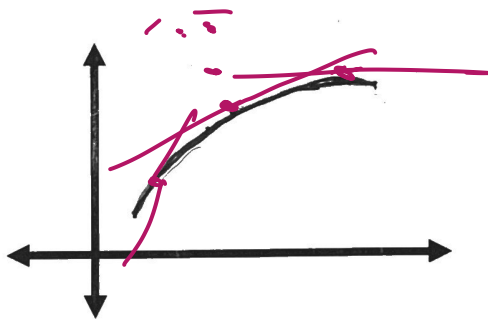
- bent so that the graph opens upward

$f''(x)$ is positive over this interval

- tangents are below the graph

$$f''(x) > 0$$

• Concave Down (C. D.)



- bent so that the graph opens downward

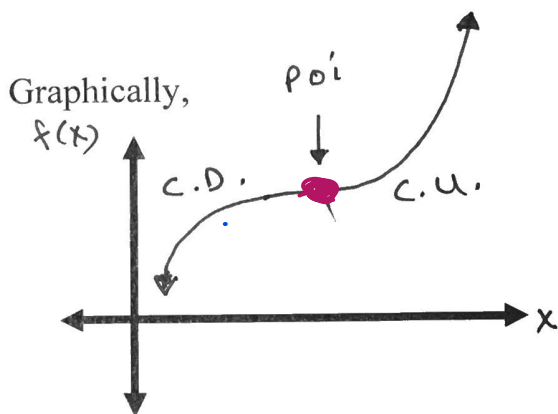
$f''(x)$ is negative over this interval

- tangents are above the graph

$$f''(x) < 0$$

Point of Inflection (P. O. I.)

The point where the graph changes from concave up to concave down or vice versa is called a P. O. I.



Note: $f''(x) = 0$ at a P. O. I.

or $f''(x)$ undefined

Test for Concavity

If $f''(x) > 0$ for all x on some interval, then the graph of $f(x)$ is concave up on that interval.

If $f''(x) < 0$ for all x on some interval, then the graph of $f(x)$ is concave down on that interval.

To Find Possible P. O. I.

Solve $f''(x) = 0$ or determine where $f''(x)$ DNE. Use an interval chart to determine on which intervals $f''(x)$ is positive or negative.

Example 1: Determine where the following curves are C. U. or C. D. and locate any P. O. I.

a) $f(x) = x^3 - 3x^2 + 4x$

$$f'(x) = 3x^2 - 6x + 4$$

$$f''(x) = 6x - 6$$

$$= 6(x-1)$$

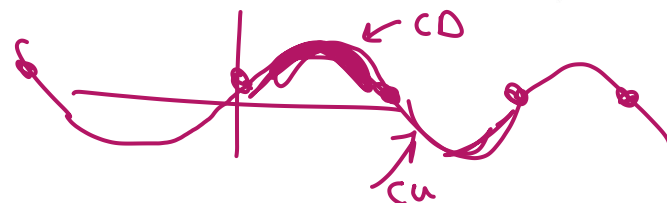
possible p.o.i.

$$f''(x) = 0$$

$$6(x-1) = 0$$

$$\therefore x = 1$$

$$y = \sin(x)$$



Interval	$(-\infty, 1)$	$(1, \infty)$
$x-1$	$-$	$+$
$f''(x)$	$-$	$+$
$f(x)$	C.D.	C.U.

$\therefore$ poi at $(1, 2)$

concave down @ $(-\infty, 1)$

concave up @ $(1, \infty)$

$$f(x) = x^{\frac{5}{2}}$$

b) $f(x) = \sqrt[3]{x} = x^{\frac{1}{3}}$

$f'(x) = \frac{1}{3} x^{-\frac{2}{3}}$

$f''(x) = -\frac{2}{9} x^{-\frac{5}{3}} = -\frac{2}{9 \sqrt[3]{x^5}}$

Vertical tangent
exponent is a fraction
denominator is odd > 1
numerator is odd
num < denom.

possible poi $f''(x) \text{ DNE @ } x = 0$

$\frac{-2}{9 x^{5/3}} \Rightarrow x = 0$

$(-8)^{5/3}$

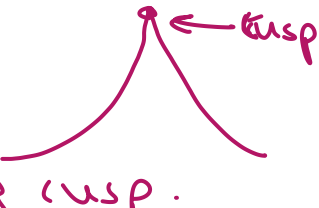
Interval	$(-\infty, 0)$	$(0, \infty)$
$-\frac{2}{9}$	—	—
$x^{5/3}$	+	+
$f''(x)$	+	—
$f(x)$	c.u.	c.d.

$\therefore$ poi @ $(0, 0)$

concave up @ $(-\infty, 0)$

concave down @ $(0, \infty)$

Cusps and Vertical Tangents

$$f'(c) = 0 \text{ or } f'(c) \text{ DNE! } \left. \begin{array}{l} \text{ } \\ \text{ } \end{array} \right\} \text{ cusp.}$$


- A cusp is a pointy bend on the graph where no unique tangent line exists.

$$f(x) = (x-3)^{\frac{1}{3}}$$

- A cusp is a good example of the situation where a function can have a maximum or a minimum at a point where the derivative of the function DNE.

- To locate a cusp, determine where $f'(x)$ DNE and then test the concavity on either side of that critical number.

$$f(x) = x^{\frac{2}{3}}$$

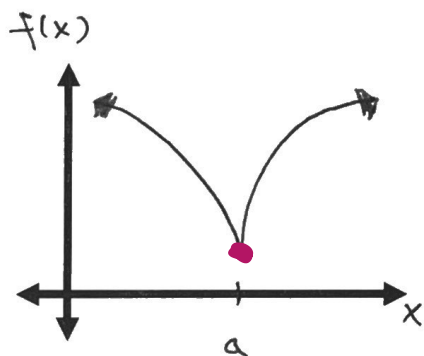
$$\Rightarrow x^{\frac{4}{3}}$$

Cusps

→ If the concavity remains the same there is a **cusp**.

→ If the concavity changes there is a **vertical tangent**.

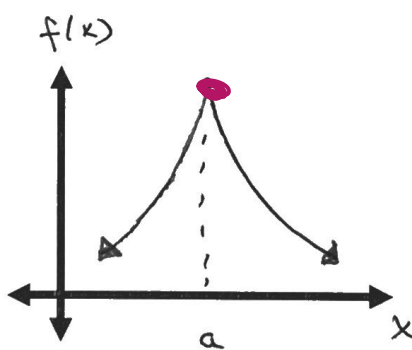
$$\begin{array}{c|c} x^{\frac{1}{3}} & \sqrt{x} \\ \hline x^{\frac{4}{3}} & \sqrt{x} \end{array}$$



* cusp at $x=a$

* No p.o.i.

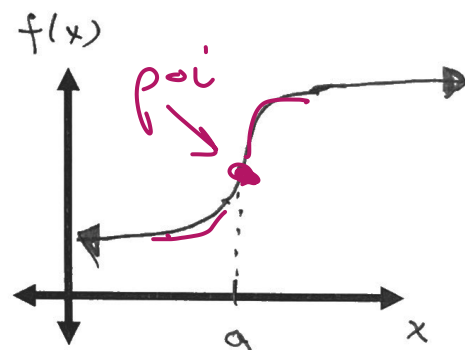
* local minimum



* cusp at $x=a$

* No p.o.i.

* local maximum



* vertical tangent at $x=a$.

* p.o.i. at the vertical tangent.

Example 2: Verify that $f(x) = x^{\frac{2}{3}}$ has a cusp at $x = 0$.

● $f'(x) = \frac{2}{3} x^{-\frac{1}{3}} = \frac{2}{3\sqrt[3]{x}}$

critical #: $f'(x) \text{ DNE @ } x = 0$

1st D.T.

Interval	$(-\infty, 0)$	$(0, \infty)$
$x^{\frac{1}{3}}$	-	+
$f'(x)$	-	+
$f(x)$	↓	↑

optional

↑ local min @ $(0, 0)$

● $f''(x) = -\frac{2}{9} x^{-\frac{4}{3}}$

possible poi

$f''(x) \text{ DNE @ } x = 0$

Interval	$(-\infty, 0)$	$(0, \infty)$
$-\frac{2}{9}$	-	-
$x^{-\frac{4}{3}}$	+	+
$f''(x)$	-	-

$f(x)$ C.D. C.D. ← no change in

● $f''(x)$ C.D. 0 C.D.

$$x^{-\frac{4}{3}} = \frac{1}{\sqrt[3]{x^4}} = \frac{1}{\sqrt[3]{(-8)^4}} = \frac{1}{16}$$

concavity

⇒ cusp @ $(0, 0)$

Test for Local Extrema using the 2nd Derivative Test

Given c is a critical number, then

→ If $f'(c) = 0$ and $f''(c) > 0$, then $f(x)$ has a local minimum at c .

→ If $f'(c) = 0$ and $f''(c) < 0$, then $f(x)$ has a local maximum at c .

→ If $f'(c) = 0$ and $f''(c) = 0$, then the 2nd Derivative Test fails.

Example 3: Use the 2nd Derivative Test to find the local extrema of $f(x) = 3x^5 - 5x^3$.

$$f'(x) = 15x^4 - 15x^2$$

$$= x^2(3x^2 - 5)$$

$$= 15x^2(x^2 - 1) = 0$$

critical #'s: $x = 0, \pm 1$

$$-60 + 30 = -30$$

$$f''(x) = \frac{60x^3 - 30x}{0}$$

$$f''(0) = 0$$

Test fails.

* use 1st derivative test!

$$f''(1) = 30 > 0$$

$\Rightarrow$ local min.

@ $(1, -2)$

$$f''(-1) = -30 < 0$$

$\Rightarrow$ local max.

@ $(-1, 2)$